

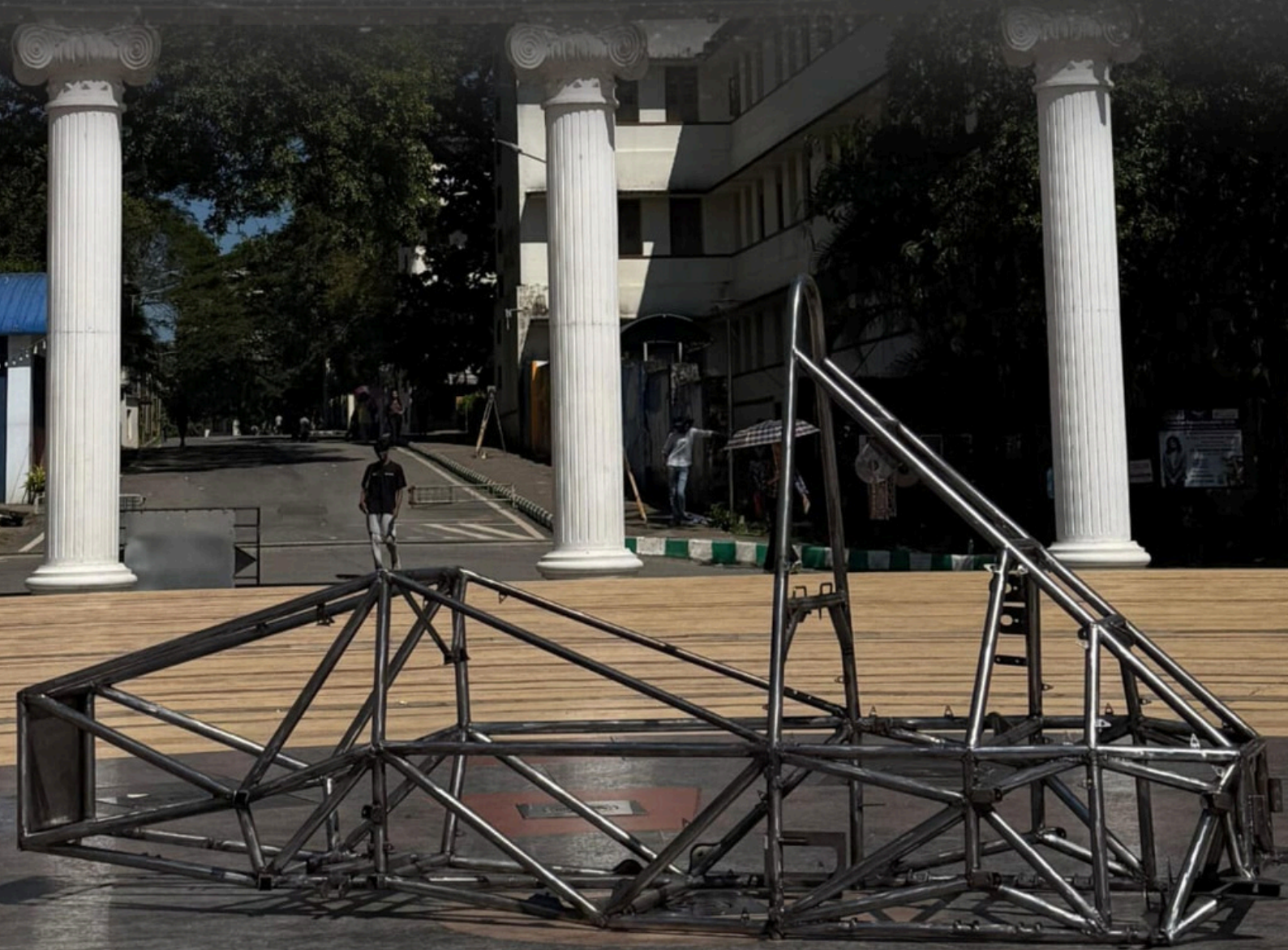
FORMULA MONTHLY

OCTOBER 2025



FORMULA
manipal

Engineered with Passion



TAKE A LOOK INTO OUR PROGRESS



Structures

The team completed key design, analysis, and manufacturing tasks, including 2D torsional analysis of multiple chassis iterations to assess deformation, stiffness, and weight, along with the development of several monocoque versions (V4-V6). These efforts were supported by ACP testing, composite insert fabrication, and validation studies. In parallel, major fabrication milestones were achieved, such as welding and profiling critical chassis brackets, rewelding TSAC brackets, manufacturing and bolting the firewall, and installing the pushjack fire extinguisher mount, while also completing essential documentation and academic task-phase activities.

Aerodynamics and composites

The Aerodynamics and Composites team completed the manufacturing of the mainplanes of the wings. Final assembly of the wings are in progress. Layups for the rear body panels were completed. Layup for the nosecone is in progress. The radiator and sidepod design, size, and position was finalized.

Vehicle dynamics

The team completed major braking and suspension milestones, including brake line fabrication, brake assembly and bleeding setup, and design and validation of the dead-man braking mechanism. Analytical work progressed with calculation of Magic Formula coefficients, suspension studies using MATLAB and OptimumK, and tensile testing of CF links, while hardware development included hub offset machining, and continued refinement of suspension and pedal system integration.

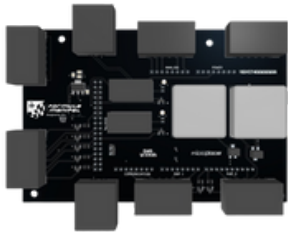


Transmission

The team completed key documentation and fabrication tasks, including vibrational analysis EDP preparation and welding of intermediate shaft and motor mount brackets. In parallel, studies on Shigley's, harmonic and force analysis, lubrication, and integration of the dual chain drive with the Emrax 208 motor progressed, supporting ongoing refinement of the transmission system

E-Powertrain

The team completed major hardware and integration tasks, including BLDC motor modeling, TSAC assembly, installation of the HVD enclosure, and welding of motor mount and TSAC brackets onto the chassis. Progress was also made on accumulator segment assembly and rewiring, motor and controller cooling circuits for static testing, and lap time simulations using OptimumLap, moving the powertrain closer to full vehicle integration

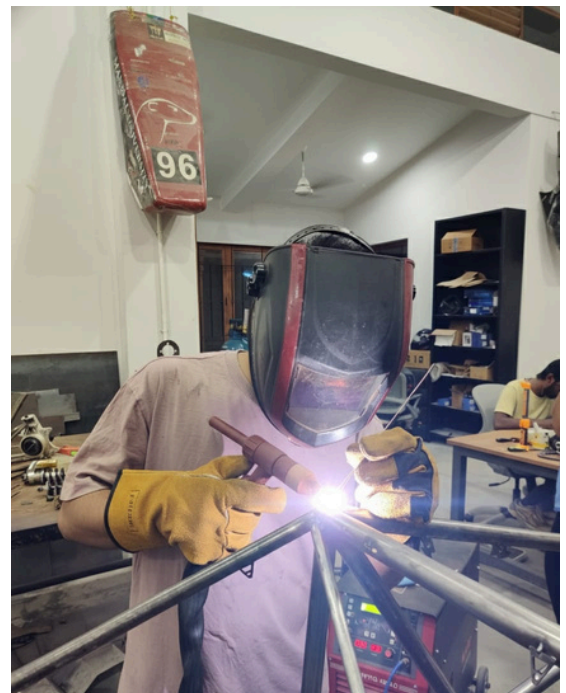


Electronics and controls

The team completed key wiring and validation tasks, including front and rear E-box wiring, shutdown wiring, load testing of the MOSFET PDU, and static testing of the car. In parallel, telemetry and data systems were advanced through ESP32-based data transmission via MQTT, storage in MongoDB, and integration with a Visual Studio dashboard, while wiring documentation and initial PCB design work were also completed to support further system refinement

Driverless

The team achieved steady progress in perception, control, and hardware integration, including validation of SBW motor functionality, refinement of LV and HV motor circuitry with working precharge and discharge, and finalization of SBW and BBW designs. On the software side, cone perception and data fusion were improved using EKF-based logic, MPC and sectoring logic were refined for FSDS, and SLAM, GNSS, and velocity estimation pipelines were actively evaluated. In parallel, key mechanical tasks such as driving sprocket design, upright installation, and preparation for SBW/BBW manufacturing were completed, moving the driverless system closer to reliable integration and testing



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